

increases the invariant with no effect on price, whereas trading against a market impacts the price with no effect on the invariant.

Each trade in a Uniswap market has an associated 0.3 percent fee that is paid back into the pool. Liquidity providers earn these fees based on their pro rata contribution to the liquidity pool and therefore prefer high-volume markets. This mechanism of earning fees is identical to the *cToken* model of Compound. The ownership stake is represented by a similar token called a UNI token. For example, the token representing ownership in the DAI/ETH pool is UNI DAI/ETH.

Liquidity providers in Uniswap essentially earn passive income in proportion to the volume on the market they are supplying. On withdrawal, however, the exchange rate of the underlying assets will almost certainly have changed. This shift creates an opportunity–cost dynamic (*impermanent loss*) that arises because the liquidity provider could simply hold the underlying assets and profit from the price movement. The fees earned from trading volume must exceed impermanent loss for liquidity providing to be profitable. Consequently, stablecoin trading pairs such as USDC/DAI are attractive for liquidity providers because the high correlation of the assets minimizes the impermanent loss.

Uniswap's $k = x * y$ pricing model works well if the correlation of the underlying assets is unknown. The model calculates the exact same slippage at a given liquidity level for any two trading pairs. In practice, however, we would expect much lower slippage for a stablecoin trading pair than for